



M2 Browning

The **M2 machine gun** or **Browning .50 caliber machine gun** (informally, "**Ma Deuce**")^{[13][14]} is a heavy machine gun that was designed near the end of World War I by John Browning. While similar to Browning's M1919 Browning machine gun, which was chambered for the .30-06 cartridge, the M2 uses Browning's larger and more powerful .50 BMG (12.7 mm) cartridge. The design has had many designations; the official U.S. military designation for the infantry type is **Browning Machine Gun, Cal. .50, M2, HB, Flexible**. It has been used against infantry, light armored vehicles, watercraft, light fortifications, and low-flying aircraft.

The gun has been used extensively as a vehicle weapon and for aircraft armament by the United States since the 1930s. It was heavily used during World War II, the Korean War, the Vietnam War, the Falklands War, the Soviet–Afghan War, the Gulf War, the Iraq War, and the War in Afghanistan. It is the primary heavy machine gun of NATO countries and has been used by many other countries as well. U.S. forces have used the M2 longer than any other firearm except the .45 ACP M1911 pistol, which was also designed by John Browning.

The **M2HB** (heavy barrel) is manufactured in the U.S. by General Dynamics,^[15] Ohio Ordnance Works,^[16] U.S. Ordnance,^[17] and FN Herstal for sale to the U.S. government and other nations via Foreign Military Sales.

History

Machine guns were heavily used in World War I, and weapons of larger than rifle caliber began appearing on both sides of the conflict.

Browning machine gun, cal. .50, M2, HB



M2E2 with a quick change barrel and tripod

Type	<u>Heavy machine gun</u>
Place of origin	<u>United States</u>
Service history	
In service	<u>1933–present</u>
Used by	See <u>Users</u>
Wars	<u>World War II</u> <u>Indonesian National Revolution</u> <u>Korean War</u> <u>First Indochina War</u> <u>Suez Crisis</u> <u>1958 Lebanon crisis</u> <u>Cuban Revolution</u> <u>Portuguese Colonial War</u> <u>Rhodesian Bush War</u> <u>Vietnam War</u> <u>Laotian Civil War</u> <u>Cambodian Civil War</u> <u>Colombian Armed Conflict</u> <u>Dominican Civil War</u> ^[1] <u>Ethiopian Civil War</u> <u>Lebanese Civil War</u> <u>Cambodian–Vietnamese War</u> <u>Sino-Vietnamese War</u> <u>Nicaraguan Revolution</u> <u>Falklands War</u> <u>Operation Urgent Fury</u> <u>Kurdish–Turkish conflict (1978–present)</u> <u>Iran–Iraq War</u> <u>Operation Just Cause</u>

The larger rounds were needed to pierce the armor that was being introduced to the battlefield, both on the ground and in the air. Germany introduced the Junkers J.I aircraft, whose armor could render ineffective aircraft machine guns that used conventional rifle ammunition such as the .30-06.^[18] Consequently, the American Expeditionary Force's commander General John J. Pershing asked for a larger-caliber machine gun.^[19] Pershing asked the Army Ordnance Department to develop a machine gun with a caliber of at least 0.50 inches (12.7 mm) and a muzzle velocity of at least 2,700 feet per second (820 m/s).^[18]

Around July 1917, John Browning started redesigning his .30-06 M1917 machine gun for a larger and more powerful round. Winchester worked on the cartridge, which was a scaled-up version of the .30-06. Winchester initially added a rim to the cartridge because the company wanted to use the cartridge in an anti-tank rifle, but Pershing insisted the cartridge be rimless.^[20] The first .50-caliber machine gun underwent trials on 15 October 1918. It fired at less than 500 rounds per minute, and the muzzle velocity was only 2,300 ft/s (700 m/s). Cartridge improvements were promised.^[21] The gun was heavy, difficult to control, fired too slowly for the anti-personnel role, and was not powerful enough against armor.^[22]

While the .50-caliber was being developed, some 13.2×92mmSR Mauser 1918 T-Gewehr anti-tank rifles and their ammunition were captured. The 13.2 mm German rounds had a muzzle velocity of 2,700 ft/s (820 m/s), an 800 gr (52 g) bullet, and could penetrate armor 1 in (25 mm) thick at a range of 250 yd (230 m).^[23] Winchester improved the .50 caliber round to have similar performance. Ultimately, the muzzle velocity was 2,750 ft/s (840 m/s).^[24]

- Persian Gulf War
- Rwandan Civil War^[2]
- Somali Civil War
- Yugoslav Wars
- Operation Uphold Democracy
- War in Afghanistan (2001–2021)^[3]
- Iraq War
- Syrian Civil War (2011–present)
- War in Iraq (2013–2017)^[4]
- Yemeni Civil War (2014–present)
- Russo-Ukrainian War^[5]

Production history	
Designer	<u>John M. Browning</u>
Designed	1918
Manufacturer	Active <u>Fabrique Nationale</u> <u>General Dynamics</u> <u>Ohio Ordnance Works Inc.</u> <u>U.S. Ordnance</u> Former <u>Manroy Engineering</u> <u>Sabre Defence Industries</u> <u>Colt's Patent Firearms Manufacturing Company</u> <u>High Standard Firearms</u> <u>Savage Arms Corporation</u> <u>Buffalo Arms Corporation</u> General Motors Corporation <u>Frigidaire</u> <u>AC Spark Plug</u> <u>Saginaw Steering Gear Division</u> <u>Brown-Lipe-Chappin Divisions</u> <u>Kelsey Hayes Wheel Company</u> <u>Springfield Armory</u> <u>Wayne Pump Company</u>

Efforts by Browning and Fred T. Moore resulted in the water-cooled, .50 caliber M1921 Browning machine gun and an aircraft version. These guns were used experimentally from 1921 until 1937. They had lightweight barrels and the ammunition fed only from the left side. Service trials raised doubts about whether the guns would be suitable for aircraft or for anti-aircraft use. A heavy barrel M1921 was considered for ground vehicles.^[25]

John M. Browning died in 1926. Between 1927 and 1932, S. H. Green studied the design problems of the M1921 and the needs of the armed services. The result was a single receiver design that could be turned into seven types of .50 caliber machine guns by using different jackets, barrels, and other components. The new receiver allowed right or left side feed. In 1933, Colt manufactured several prototype Browning machine guns (including what would be known as the M1921A1 and M1921E2). With support from the Navy, Colt started manufacturing the M2 in 1933.^[26] FN Herstal (Fabrique Nationale) has manufactured the M2 machine gun since the 1930s.^[27] General Dynamics, U.S. Ordnance and Ohio Ordnance Works Inc. are other current manufacturers.

A variant without a water jacket, but with a thicker-walled, air-cooled barrel was designated the M2 HB (*HB* for *Heavy Barrel*). The added mass and surface area of the heavy barrel compensated somewhat for the loss of water-cooling, while reducing bulk and weight: the M2 weighs 121 lb (55 kg) with a water jacket, but the M2 HB weighs 84 lb (38 kg). Due to the long procedure for changing the barrel, an improved system was developed called QCB (quick change barrel). The lightweight "Army/Navy" prefixed **AN/M2**

"light-barrel" version of the Browning M2 weighing 60 pounds (27 kg) was also developed, and became the standard .50-caliber aviation machine gun of the World War II-era for American military aircraft of nearly every type,^[28] readily replacing Browning's own air-cooled .30 caliber machine gun design in nearly all American aircraft installations.

	ERMCO
	<u>Ramo Manufacturing</u>
	<u>Rock Island Arsenal</u>
Produced	1921–present (M2HB/M2A1)
<u>No. built</u>	3 million ^[6]
Specifications	
Mass	38 kg (84 lb) 28 kg (62 lb) (AN/M2) 58 kg (128 lb) with tripod and traverse and elevation mechanism (T&E) 24 lb (11 kg) barrel weight^[7]
Length	1,654 mm (65.1 in) 1,429 mm (56.3 in) (AN/M2)
<u>Barrel length</u>	1,143 mm (45.0 in) 910 mm (35.8 in) (AN/M2)
<u>Cartridge</u>	<u>.50 BMG (12.7×99mm NATO)</u>
<u>Action</u>	<u>Short recoil-operated</u>
<u>Rate of fire</u>	450–600 rounds/min (M2HB) ^{[8][9]} 750–850 rounds/min (AN/M2) 1,200–1,300 rounds/min (AN/M3) ^[10]
<u>Muzzle velocity</u>	2,910 ft/s (890 m/s) (M33 ball), 78 ft (24 m) from muzzle ^[11] 3,050 ft/s (930 m/s) in manual ^{[7][12]}
<u>Effective firing range</u>	1,800 m (2,000 yd) ^[8]
<u>Maximum firing range</u>	7,400 m (8,100 yd)
<u>Feed system</u>	<u>Belt-fed</u> (M2 or M9 links)

Design details

The Browning M2 is an air-cooled, belt-fed machine gun. The M2 fires from a closed bolt, operated on the short recoil principle. The M2 fires the .50 BMG cartridge, which offers longer range, greater accuracy, and immense stopping power. The closed bolt firing cycle made the M2 usable as a synchronized machine gun on aircraft before and during World War II, as on the early versions of the Curtiss P-40 fighter. The M2 is a scaled-up version of John Browning's M1917 .30 caliber machine gun.

Features

The M2 has varying cyclic rates of fire, depending on the model. The M2HB air-cooled ground gun has a cyclical rate of 450–575 rounds per minute.^[29] The early M2 water-cooled AA guns had a cyclical rate of around 450–600 rpm.^[30] The AN/M2 aircraft gun has a cyclic rate of 750–850 rpm; this increases to 1,200 rpm for AN/M3 aircraft guns. These maximum rates of fire are generally not achieved in use, as sustained fire at that rate will wear out the bore within a few thousand rounds, necessitating replacement. In addition to full automatic, the M2HB can be selected to fire single shots, fire slowly at less than 40 rounds per minute, or fire rapidly for more than 40 rounds per minute. Slow and rapid firing modes use 5–7 round bursts with different lengths of pause between bursts.^[31]



A U.S. Marine mans a .50 caliber machine gun as part of a security force during a training exercise with the 24th Marine Expeditionary Unit in November 2002.

The M2 has an effective range of 1,830 metres (2,000 yd) and a maximum effective range of 2,000 metres (2,200 yd) when fired from the M3 tripod. In its ground-portable, crew-served role as the M2HB, the gun itself weighs 84 pounds (38 kg) and the assembled M3 tripod another 44 pounds (20 kg). In this configuration, the V-shaped "butterfly" trigger is located at the very rear of the weapon with a "spade handle" handgrip on either side of it and the bolt release in the center. The spade handles are gripped and the butterfly trigger is depressed with one or both thumbs. Recently, new rear buffer assemblies have used squeeze triggers mounted to the handgrips, doing away with the butterfly triggers.

When the bolt release is locked down by the bolt latch release lock on the buffer tube sleeve, the gun functions in fully automatic mode. Conversely, the bolt release can be unlocked into the up position resulting in single-shot firing (the gunner must press the bolt latch release to send the bolt forward). Unlike virtually all other modern machine guns, it has no safety (although a sliding safety switch has recently been fielded to USMC armorers for installation on their weapons and is standard-issue for the U.S. Army for all M2s). Troops in the field have been known to add an improvised safety measure against accidental firing by slipping an expended shell casing under the butterfly trigger.^[32] The upgraded M2A1 has a manual trigger block safety.

Because the M2 was designed to operate in many configurations, it can be adapted to feed from the left or right side of the weapon by exchanging the belt-holding pawls, and the front and rear cartridge stops (three-piece set to include link stripper), then reversing the bolt switch. The operator must also convert

the top-cover belt feed slide assembly from left to right-hand feed as well as the spring and plunger in the feed arm. This will take a well-trained individual less than two minutes to perform.

The charging assembly may be changed from left to right-hand charge. A right-hand charging handle spring, lock wire, and a little "know-how" are all that is required to accomplish this. The M2 can be battle-ready and easily interchanged if it is preemptively fitted with a retracting slide assembly on both sides of the weapon system. This eliminates the need to have the weapon removed from service to accomplish this task.

At some point during World War II, the Frankford Arsenal developed a squeeze bore version of the M2HB which reduced the bullet size from .50 to .30 caliber.^[33]

Ammunition

There are several different types of ammunition used in the M2HB and AN aircraft guns. From World War II through the Vietnam War, the Browning was used with standard ball, armor-piercing (AP), armor-piercing incendiary (API), and armor-piercing incendiary tracer (APIT) rounds. All .50 ammunition designated "armor-piercing" was required to completely perforate 0.875 inches (22.2 mm) of hardened steel armor plate at a distance of 100 yards (91 m) and 0.75 inches (19 mm) at 547 yards (500 m).^[34] The API and APIT rounds left a flash, report, and smoke on contact, useful in detecting strikes on enemy targets; they were primarily intended to incapacitate thin-skinned and lightly armored vehicles and aircraft, while igniting their fuel tanks.^[35]

Current ammunition types include M33 Ball (706.7 grain) for personnel and light material targets, M17 tracer, M8 API (622.5 grain), M20 API-T (619 grain), and M962 SLAP-T. The latter ammunition along with the M903 SLAP (Saboted Light Armor Penetrator) round can perforate 1.34 inches (34 mm) of FHA (face-hardened steel plate) at 500 metres (550 yd), 0.91 inches (23 mm) at 1,200 metres (1,300 yd), and 0.75 inches (19 mm) at 1,500 metres (1,600 yd). This is achieved by using a 0.30-inch-diameter (7.6 mm) tungsten penetrator. The SLAP-T adds a tracer charge to the base of the ammunition. This ammunition was type classified in 1993.

When firing blanks, a large blank-firing adapter (BFA) of a special type must be used to allow the recoil-operated action to cycle. This functions on the principle of a recoil booster, to increase the recoil force acting on the short recoil action. This is the exact antithesis of a muzzle brake. Without this adaptor, the reduced-charge blank cartridge would develop too little recoil to cycle the action fully. The adapter is very distinctive, attaching to the muzzle with three rods extending back to the base. The BFA can often be seen on M2s during peacetime operations.



Twin M2HB machine gun during a pre-action calibration fire (PACFIRE) exercise in May 2005

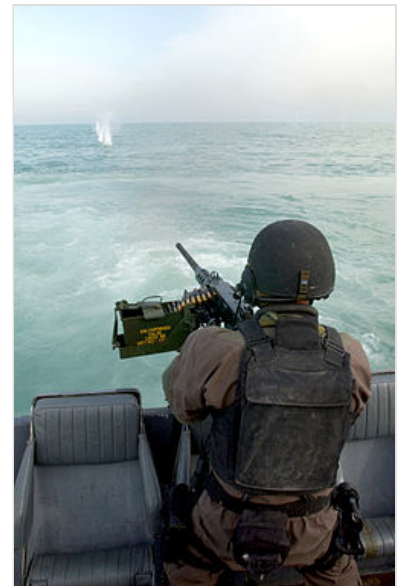


M2 with blank-firing adapter

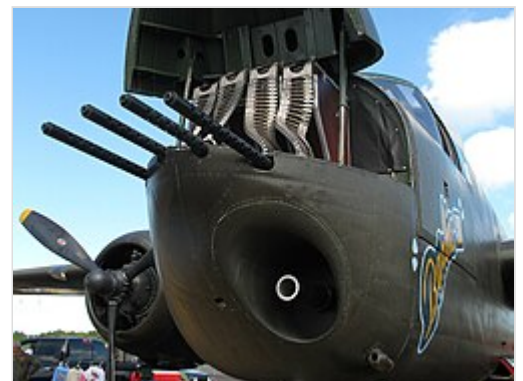
Deployment

The M2 .50 Browning machine gun has been used for various roles:

- A medium infantry support weapon
- As a light anti-aircraft (AA) gun in some ships; up to six M2 guns could be mounted on the same turret.
- As an anti-aircraft gun on the ground. The original water-cooled version of the M2 was used on a tall AA tripod or vehicle-mounted anti-aircraft weapon on pedestal mount. In later variants, twin and quadruple M2HB Brownings were used, such as the M45 Quadmount (aka "meat chopper") used on the US M16 half-track carrier. Twin or quad-mount .50 M2 guns normally used alternating left-hand and right-hand feed.
- Primary or secondary weapon on an armored fighting vehicle.
- Primary or secondary weapon on a naval patrol boat.
- Spotting for the primary weapon on some armored fighting vehicles.
- Secondary weapon for anti-boat defense on large naval vessels (corvettes, frigates, destroyers, cruisers, etc.).
- Coaxial gun or independent mounting in some tanks, including but not limited to: the M47 Patton, M48 Patton, M4 Sherman, M24 Chaffee, Heavy tank M6, Heavy Tank T29, M1 Abrams, M60 Patton, M46 Patton, and the M26 Pershing.
- Fixed-mounted forward-firing primary aircraft armament (AN/M2 and AN/M3 light-barrel versions only). The AN/M2 was used as primary armament in almost all World War II U.S. pursuit aircraft (such as the North American P-51 Mustang, Republic P-47 Thunderbolt, Lockheed P-38 Lightning, Bell P-39 Airacobra, Curtiss P-40 Warhawk, Grumman F6F Hellcat, and Vought F4U Corsair). It was also used in fixed mountings in bombers and ground attack aircraft like the Douglas SBD Dauntless dive bomber, Grumman TBF Avenger torpedo bomber, and medium bombers such as North American B-25 Mitchell, Martin B-26 Marauder, and Douglas A-26 Invader; usually 4–8 per aircraft but the bombers could mount 12 or more in certain configurations. The later, faster-firing electrically feed-boostered AN/M3 was used in many Korean War–era USAF fighter aircraft such as the Lockheed F-80 Shooting Star, Republic F-84 Thunderjet, North American F-86 Sabre, and early versions of the Martin B-57 Canberra bomber. The US Navy had largely completed their move to the (unrelated) M2/AN 20 mm autocannon for aircraft armament by this time.
- Turret-mount or flexible-mounted defensive armament, again only with the AN/M2 light-barrel version, in almost all US World War II–era bombers and patrol aircraft such as the Boeing B-17 Flying Fortress, Consolidated B-24 Liberator and Boeing B-29 Superfortress heavy bombers, North American B-25 Mitchell and Martin B-26 Marauder medium bombers, Consolidated PBV Catalina patrol flying boats, Goodyear K- and M-class blimps, Grumman TBF/TBM Avenger torpedo bombers, and in a combined offensive/defensive turret mounting



An M2 fired from a rigid-hulled inflatable boat.



B-25H "Barbie III" showing four M2 feeds and 75 mm M5 gun

in many Northrop P-61 Black Widow night fighters. The AN/M3 was used as a flexible, quad-mounted, radar-directed tail-defense gun as late as 1980 on the Boeing B-52 Stratofortress, until replaced by 20 mm M61 Vulcan Gatling-type cannon on the H model.^[36]

- Variants of the AN/M3 are used as flexible door guns or as flexible remotely-controlled armament subsystems on many US Army, Navy, Marine Corps and Coast Guard helicopters, such as the Bell UH-1 Iroquois, Sikorsky UH-60 Blackhawk and variants, Sikorsky CH-53E Super Stallion, Bell OH-58 Kiowa, and others.

United States

At the outbreak of the Second World War, the United States had versions of the M2 in service as fixed aircraft guns, anti-aircraft defensive guns (on aircraft, ships, or boats), infantry (tripod-mounted) guns, and as dual purpose anti-aircraft and anti-vehicular weapons on vehicles.^{[37][38]}

The .50 AN/M2 light-barrel aircraft Browning used in planes had a rate of fire of approximately 800 rounds per minute and was used singly or in groups of up to eight guns for aircraft ranging from the P-47 Thunderbolt to the B-25 Mitchell bomber, which in the last J-version of the Mitchell could have up to fourteen M2s firing forward for ground attack missions – eight in a solid metal-structure nose, four more mounted in a pair of conformal twin-gunned gun pods on the lower cockpit sides, and two more if the forward dorsal turret's pair of M2 guns were also aimed straight forward. The later A-26 bested this with up to a maximum of 16/18 machine guns, 8 in the nose, four more per wing in flush-mount pods, plus 2 guns in the dorsal turret.



A U.S. soldier in Normandy stands guard with the M2HB installed on a dual-purpose mounting, 1944.

In the dual-purpose vehicle mount, the M2HB proved extremely effective in U.S. service: the Browning's .50 caliber AP and API rounds could easily penetrate the engine block or fuel tanks of a German Bf 109 fighter attacking at low altitude,^[39] or perforate the hull plates and fuel tanks of a German half-track or light armored car. It could even penetrate the sides and rear of the Panzer I, Panzer II, Panzer III, and Panzer IV tanks.^{[34][40][41]} While the dual-purpose mounting was undeniably useful, it did normally require the operator to stand when using the M2 in a ground role, exposing him to return fire.^[42] Units in the field often modified the mountings on their vehicles, especially tanks and tank destroyers, to provide more operator protection in the anti-vehicular and anti-personnel role.^[43] The weapon was particularly hated by the Germans, whose attacks and ambushes against otherwise helpless stalled motor convoys were frequently broken up by .50 caliber machine gun fire.^{[44][45]} Vehicles would frequently "recon by fire" with the M2 Browning, i.e. they would fire continuously at suspected points of ambush while moving through areas still containing enemy forces. One vehicle would fire exclusively to the right, the following vehicle to the left, the next one to the right, and so on in order to cover both flanks of the advancing convoy.

Besides vehicle-mounted weapons, the heavy weapons companies in a World War II U.S. Army infantry battalion or regiment were each issued one M2 Browning with tripod (ground) mount.^[46] Mounted on a heavily sandbagged tripod, the M2HB proved very useful in either a defensive role or to interdict or block road intersections from use by German infantry and motorized forces.^[47] Hearing the sound of an M2

could often cause enemy infantry to take cover.^[48] There are numerous instances of the M2 Browning being used against enemy personnel, particularly infantry assaults^[49] or for interdiction or elimination of enemy artillery observers or snipers at distances too great for ordinary infantry weapons.^{[50][51][52]}

The M2HB was not widely used in the Pacific campaign for several reasons, including the weight of the gun, the nature of infantry jungle combat, and because road intersections were usually easily outflanked.^[53] However, it was used by fast-moving motorized forces in the Philippines to destroy Japanese blocking units on the advance to Manila.^[47] The quad mount .50 was also used to destroy Japanese emplacements.^[54]



An M2 overlooking the Korengal Valley at Firebase Phoenix, Afghanistan, in 2007

The M2HB was used in Korea and Vietnam, and later in both Operation Desert Storm, the Afghan theater of Operation Enduring Freedom and in Iraq. In 2003, U.S. Army SFC Paul Ray Smith used his M2HB mounted on an M113 armored personnel carrier to kill 20 to 50 enemies who were attacking a U.S. outpost, preventing an aid station from being overrun and allowing wounded soldiers to be evacuated,^[55] SFC Smith was killed during the firefight and was posthumously awarded the Medal of Honor.

M45 Quadmount

The M45 Quadmount was a mounting of four .50 M2HB guns with a single gunner situated behind an armored housing. This was used by U.S. anti-air battalions, fitted either on a towed trailer or mounted on a half-track carrier. With 200 rounds per gun in a powered tracking mount, the guns proved very effective against low-flying aircraft. The use of four guns adequately compensated for the fact that the individual M2HB's rate of fire (450–550 rounds per minute) was low for an effective anti-aircraft weapon.^[56]



An M45 Quadmount installed on an M16 Multiple Gun Motor Carriage

Towards the end of the war, as Luftwaffe attacks became less frequent, the quad .50 (nicknamed the *Meat Chopper* or *Krautmower*^[56]) was increasingly used in an anti-personnel role, similarly to the earlier-introduced (1940) and more powerful—but much more difficult to keep well-fed with ammunition when in action—German 20 mm Flakvierling. Snipers firing from trees were engaged by the quad gunner at trunk level; the weapon would cut down and destroy the entire tree, and the sniper with it.^{[48][54]}

The M45 Quadmount was still in use during the Vietnam War.

Commonwealth and other forces

The Commonwealth use of the M2 Browning .50 caliber machine gun (known as the .5 Browning in British and Commonwealth service) began in World War II, though from 1942 it was standard armament on US-built AFVs provided under lend-lease such as the M4 Sherman, M7 Priest, M8 Greyhound, or

M10 tank destroyer variously used by British, Canadian, Australian, South African, and New Zealand units. Nevertheless, the heavy Browning's effectiveness was praised by many British and Commonwealth soldiers in infantry, armored, and ordnance branches.^{[57][58]} Many commanders thought that the .50 Browning was the best weapon in its class, certainly the best of the American weapons, including the M1 Garand and M1 Carbine.^{[58][59]} In North Africa, after Commonwealth units began to obtain sufficient parts, manuals, gauges, and ammunition for the new weapon, the .50 Browning was increasingly used, eventually replacing the 15 mm Besa,^[58] but in Italy it was often deleted from top turret mountings because the mount exposed the operator to low branches and enemy fire.^[60] All LRDGs, and some SAS units used the aircraft (AN/M2) version of the gun, while beam/waist-mounted and turret-mounted Brownings were used later in the war in such aircraft as the Short Sunderland and Lancaster bomber.

After World War II, the .50 Browning continued to see action in Korea and other theaters, in aircraft, tripod (ground), ground AA (hip-ring), and vehicle mounts. One of its most notable actions in a ground role was in a fierce battle with a nine-man Special Air Service team at the Battle of Mirbat in Oman in July 1972, where the heavy Browning and its API ammunition was used to help repulse an assault by 250 Yemeni Adoo guerrillas, though the more famous weapon from the battle is a 25 pounder gun.^[61] The Scots Guards used the weapon in the 1982 Falklands War.^[62]

A .50 caliber Browning was installed along with a .30 caliber Browning machine gun in each compact one-man turret on M113 APCs used by the Royal Australian Armoured Corps in South Vietnam.

The M2HB has been in service with the Israel Defense Forces since its establishment and has served in all of Israel's wars, operations, and conflicts. In 2012, the IDF upgraded its M2HB machine guns to the M2HQCB model, with a heavy quick-change barrel. Today the M2 serves as an infantry crew-served heavy machine gun, as a remote-controlled external coaxial gun on Merkava main battle tanks, as the main weapon on the Samson RCWS, and as a secondary weapon on Israeli Sea Corps gunboats and missile boats.

Nigerian troops have extensively deployed the 50 caliber Browning, mounted on Otokar Cobra APCs, Panhard VBL M11s and Landcruiser gun-trucks in counterinsurgency operations in the Niger Delta, N.E Nigeria, the Jos Plateau, and in Mali.



Australian M113 with twin mounted M1919 Browning and M2 Browning Quick Change Barrel machine guns, 17 May 2002



Israel Defense Forces M2HQCB

Japanese Usage

Use by the Imperial Japanese Military

During the Pacific War, the Empire of Japan, engaged in combat with the United States, made extensive use of M2 Browning-style machine guns and their modified versions, including fixed and swivel-mounted aircraft guns, primarily by the Imperial Army's air forces.

- Ho-103: The Imperial Japanese Army adopted the Type 103 (Ishiki Jūni-7mm Kōkū Kikō) machine gun, a modified version of the AN/M2 (MG53-2) aircraft-mounted variant of the M2 Browning, changing the caliber to 12.7x81mmSR to match the Breda SAFAT 12.7mm heavy machine gun's standard. This weapon was widely equipped on Army fighter planes, including the early to mid-period Pacific War aircraft like the Mitsubishi A6M Zero. Compared to the M2, the Ho-103 was smaller, lighter, and had a higher rate of fire, but its reduced projectile weight meant it had lower power and muzzle velocity (though it could fire high-explosive “Ma-dan” shells, a feature the M2 lacked). While the M2 was relatively large and heavy for an aircraft gun, the Ho-103 succeeded in keeping weight down.



Foreground: Type 97 Ho-5 20mm Fixed Aircraft Cannon, Center: Type 103 Ho-103 12.7mm Fixed Aircraft Cannon (Stored at the Royal Thai Air Force Museum)

- Ho-5: The Army also pursued the development of a higher-powered 20mm cannon, which led to the creation of the Ho-5 (Type 2 20mm Fixed Aircraft Cannon), a 20mm version based on the Ho-103 design. The Ho-5 was equipped on later-stage Army fighters such as the Kawanishi N1K-J Shiden. The Ho-5 was nearly the same size and weight as the M2, despite its larger caliber, and offered excellent rate of fire, muzzle velocity, and devastating power.
- Type 3 13mm Fixed Machine Gun: The Imperial Japanese Navy also adopted a variant based on the M2, the Type 3 13mm Fixed Machine Gun, which used a 13mm cartridge (13.2x96mm) and the barrel of the Type 93 13mm heavy machine gun from the Hotchkiss family. This was mounted on a small number of Navy fighter planes, such as the A6M5b Zero, which appeared in the late stages of World War II. While the Type 3 had similar size and weight to the M2, it had superior rate of fire and a heavier bullet, giving it higher firepower per shot. However, its muzzle velocity was lower compared to the M2.

Post-World War II Use

After World War II, as Japan began rearming, the M2 was initially supplied by the United States. In addition to these supplied units, Sumitomo Heavy Industries began license production at its Tanashi factory in 1984.

In the Japan Self-Defense Forces, the 12.7mm Heavy Machine Gun M2 (12.7mm 重機関銃 M2) is primarily used in vehicles such as tanks, self-propelled guns, and armored vehicles, as well as for anti-aircraft purposes. New procurement includes approximately 80 units annually, and the M3 mount is compatible with the Type 96 40mm automatic grenade launcher. When deployed on the ground as anti-aircraft weapons, the M63 anti-aircraft mount is used. The procurement price is approximately ¥5.3 million.^[63]

The Japan Maritime Self-Defense Force initially equipped early destroyers and patrol vessels with several M2 machine guns. However, these were eventually phased out due to concerns over their insufficient power and short range. In response to incidents involving North Korean suspicious ships, the M2 machine guns were reintroduced on some vessels as more appropriate firepower for engaging small targets. The M2 is not considered a standard weapon on ships but is treated as an installed weapon (see ja:海上自衛隊の個人装備)

The Japan Air Self-Defense Force also used the M2 machine gun in the M55 trailer-mounted anti-aircraft gun for base defense purposes. However, with the introduction of newer systems like the VADS, the M2 is no longer in active combat use and is stored as a reserve.^[64]

The Japan Coast Guard has also used the M2 since its inception, referring to it as the 13mm Machine Gun and equipping many patrol boats with it. The weapon is still in use on patrol ships and boats today.

In December 2013, Sumitomo Heavy Industries was found to have falsified test data for at least 5,000 units of various machine guns, including the 5.56mm Machine Gun MINIMI, Type 74 Vehicle Mounted 7.62mm Machine Gun and the M2HB, and was placed under a five-month suspension.^[65]

In April 2021, Sumitomo Heavy Industries announced it would cease production of machine guns, although it intends to continue manufacturing parts for maintenance and repairs.^{[66][67][68]}



JGSDF soldiers firing 12.7mm Heavy Machine M2 with M63 tripod for Anti-Air purposes



another side of view of JGSDF firing M2



M2 mounted on a Type 82 command and communication vehicle belonging to the JGSDF 6th Antiaircraft Artillery Battalion

Sniper rifle

The M2 machine gun has also been used as a long-range sniper rifle when equipped with a high-powered telescopic sight. Soldiers during the Korean War used scoped M2s in the role of a sniper rifle, but the practice was most notably used by US Marine Corps sniper Carlos Hathcock during the Vietnam War. Using an Unertl telescopic sight and a mounting bracket of his own design, Hathcock could quickly convert the M2 into a sniper rifle, using the traversing-and-elevating (T&E) mechanism attached to the tripod. When firing semi-automatically, Hathcock hit man-size targets beyond 1,800 metres (2,000 yd)—twice the range of the standard-caliber sniper rifle of the time (a .30-06 Winchester Model 70). Hathcock set the record for the longest confirmed kill at 2,250 metres (2,460 yd), a record which stood until 2002, when it was broken in Afghanistan by Canadian Army sniper Arron Perry.^[69]



USMC M2 fitted with a Leupold CQBSS variable power scope

Variants and derivatives

Manufacturers

Licensed historic production:^{[70][71]}

- Aircraft Armaments Incorporated (AAI)
- Australian Defence Industries
- Canadian Arsenals Limited (CAL)
- Colt's Patent Fire Arms Company
- Buffalo Arms Company
- three divisions of General Motors: AC Sparkplug; Frigidaire; and Brown-Lipe-Chapin^[72]
- High Standard Manufacturing Company
- Kelsey Hayes Wheel Company
- Saco-Lowell Shops
- Savage Arms Corporation
- Springfield Armory (US arsenal), and
- Sumitomo Heavy Industries
- Wayne Pump Company

Current production (2023):

- Canik Arms^[73]
- FN Herstal (Fabrique Nationale)
- General Dynamics
- US Ordnance
- Ohio Ordnance Works Inc, and
- Manroy Engineering (UK)

Variants

The basic M2 was deployed in U.S. service in a number of subvariants, all with separate complete designations as per the US Army system. The basic designation as mentioned in the introduction is Browning Machine Gun, Cal. .50, M2, with others as described below.

The development of the M1921 water-cooled machine gun which led to the M2, meant that the initial M2s were, in fact, water-cooled. These weapons were designated Browning Machine Gun, Cal. .50, M2, Water-Cooled, Flexible. There was no fixed water-cooled version.

Improved air-cooled heavy barrel versions came in three subtypes. The basic infantry model, Browning Machine Gun, Cal. .50, M2, HB, Flexible, a fixed developed for use on the M6 Heavy Tank designated Browning Machine Gun, Cal. .50, M2, HB, Fixed, and a "turret type" whereby "Flexible" M2s were modified slightly for use in tank turrets. The subvariant designation Browning Machine Gun, Cal. .50, M2, HB, TT was only used for manufacturing, supply, and administration identification and separation from flexible M2s.

A number of additional subvariants were developed after the end of World War II. The M2 Heavy Barrel, M48 Turret Type was developed for the commander's cupola on the M48 Patton tank. The cupola mount on the M48A2 and M48A3 was thoroughly disliked by most tankers, as it proved unreliable in service.^[74] An externally mounted M2 was later adopted for the commander's position on the M1 Abrams tanks. Three subvariants were also developed for use by the U.S. Navy on a variety of ships and watercraft including the soft mount and fixed type versions. The fixed types fire from a solenoid trigger and come in left- or right-hand feed variants for use on the Mk 56 Mod 0 dual mount and other mounts.

Huaqing Machinery has made a clone of the M2HB known as the CS/LM6, which was released publicly in 2010 at foreign weapons expo conventions.^[75] It was made with a picatinny rail on the receiver in order to have quick installation of various optics.^[75] The original M2HB tripod and parts can be used on the CS/LM6.^[76]

M2A1

When the M2 was first being designed, John Browning faced two design challenges. With the machine tools available at that time, the dimensions that established the location of the bolt face and the depth of the chamber could not be held tightly enough to control the fit of the cartridge in the chamber. The round can be too tight in the chamber and the gun would not fire, or be too loose in the chamber, resulting in a stoppage or ruptured cartridge. The other dimension that could not be held closely enough was when the firing pin would fall. The solution to these problems was adjustable timing and headspace ("Timing" is the adjustment of the gun so that firing takes place when the recoiling parts are in the correct position for firing; "headspace" is the distance between the face of the bolt and the base of the cartridge case, fully seated in the chamber); the operator had to screw the barrel into the barrel extension, moving the barrel toward the bolt face to reach the proper headspace with simple gauges to allow the operator to adjust to the proper dimensions. By the late 20th century, the M2 was the only adjustable headspace weapon in the U.S. inventory. With rising reports of injuries from improperly headspaced weapons, the U.S. military held a competition for a quick change barrel conversion kit with fixed timing and headspace in 1997. Three companies offered kits and Saco Defense won the competition. However, funding was lost before the design could be fully evaluated and the program ended. In 2007, the military found money to start a new competition. Saco Defense had since been acquired by General Dynamics, which won the competition.^[77]

On 15 October 2010, the M2A1 heavy machine gun was type classified by the U.S. Army. Formerly known as the M2E2, the M2A1 incorporates improvements to the design including a quick change barrel (QCB) with a removable carrying handle, a new slotted flash suppressor that reduces muzzle flash by 95



An M2HB in the French Foreign Legion's 2nd Infantry Regiment during an exercise



Jungle Infantry soldier of the Brazilian Army manning an M2HB in the Amazon rainforest



M2E2 modification with quick-change barrel (QCB)

percent, fixed headspace and timing, a modified bolt, and a manual trigger block safety. When a standard M2 had a barrel change, the headspace and timing had to be manually set. Improper adjustment could damage the weapon and cause serious injury to the user. Fixed headspace and timing reduces risk, and the carrying handle allows the barrel to be switched in seconds.^{[78][79]} In June 2011, the Army began conversion of M2HB machine guns to M2A1s.^[80] The M2A1 was named one of the greatest Army inventions of 2011.^[81] As of 30 November 2012, 8,300 built or converted M2A1s had been fielded by the U.S. Army; the program will upgrade the Army's entire M2 inventory of more than 54,000 guns.^[82] The U.S. Marine Corps plans to upgrade all of their ground-mounted M2s to M2A1 standard from 2016 to 2018.^[83] The first phase of conversions was completed in March 2017, with 3,600 M2A1s planned to be fielded by the Marines in total.^[84] The Israel Defense Forces adopted the M2-HQCB (the commercial version of the M2A1) in 2012 as a replacement to the M2HB.^[85]



U.S. Army Ranger Special Operations Vehicle armed with RAMO M2HB-QCB machine gun

13,2 mm FN Browning

One derivative of the M2 Browning is the Mitrailleuse d'Avion Browning - F.N. Calibre 13,2 mm, more commonly known as the FN Browning M.1939. The FN Browning M.1939 was a heavily modified M2 Browning for aircraft use designed by FN Herstal for export. Their aim was to make a light, reliable heavy machine gun with the same damage output as a 20 mm autocannon. To achieve this, they raised the firing rate to 1080 rpm and gave it a more powerful cartridge in the form of the 13.2×99mm Hotchkiss.^[86] This cartridge was a popular Eurasian analog of .50 BMG developed independently in 1920s with a 13.2 mm bullet (.52 in) and more propellant. A new projectile was designed for this gun. It was of a high explosive type and was designed to take down a small aircraft with a single hit. Tests showed that it was very effective against both cloth and aluminum skinned aircraft.^[87]

Due to the aforementioned improvements, the gun received interest from numerous nations when it entered the export market in 1939. Due to the start of World War 2 and the invasion of Belgium it was exported to only Romania and Sweden.^[88] Sweden was able to buy the majority of the weapons along with the blueprints to produce the weapon on their own without paying for a license. In Sweden, the weapon received the designation Automatkanon m/39, short Akan m/39, meaning Autocannon m/39 and was later produced by Ericsson as the Akan m/39A. Sweden also gave the blueprints to Finland so they could produce the weapon. Since Finland was already producing 12.7 mm ammunition, the Finnish variant was rechambered to 12.7 mm (.50 BMG). The Finnish variant was designated VKT 12,70 LKk/42 and was produced by Finland.

Aircraft guns

.50 Browning AN/M2

The M2 heavy machine gun was widely used during World War II, and in later postwar conflicts, as a remote or flexible aircraft gun. For fixed (offensive) or flexible (defensive) guns used in aircraft, a dedicated M2 version was developed called the ".50 Browning AN/M2" or the "12.7 mm AN/M2".^[88] The "AN" stands for "Army/Navy", since the gun was developed jointly for use by both services. The AN/M2 designation was also used for other aircraft guns and therefore it is important to write the caliber before the designation.

The 12.7 mm AN/M2 had a cyclic rate of 600–800 rounds per minute, with the ability to be fired from an electrically operated remote-mount solenoid trigger when installed as a fixed gun.^[88] Cooled by the aircraft's slip-stream, the air-cooled 12.7 mm AN/M2 was fitted with a substantially lighter 36-inch (91 cm) length barrel, reducing the weight of the complete unit to 61 pounds (28 kg),^{[88][89]} which also had the effect of increasing the rate of fire. The full official designation for this weapon was "Browning Machine Gun, Aircraft, Cal. .50, AN/M2" (Fixed) or (Flexible).

The 12.7 mm AN/M2 was used on many aircraft during WWII, as it served as the main aircraft armament in the US military. Most US fighter aircraft were armed with four, six or eight AN/M2 MGs mounted in the wings. Some examples being the Curtiss P-40, Republic P-47 Thunderbolt, and North American P-51 Mustang for the USAAF and the F4F, F6F, and Vought F4U Corsair for the US Navy. For bombers, the AN/M2 was used in both flexible and fixed positions for both offensive and defensive use. In flexible defensive positions, the Boeing B-17G Flying Fortress heavy bomber was armed with 13 AN/M2 guns in both turreted and flexible positions. In fixed offensive configurations, like on the North American B-25 Mitchell, commonly carried 6 to 12 fixed guns for strafing.

In foreign use the AN/M2 is often just referred to as the M2 Browning. In Sweden it was re-designated **12,7 mm automatkanon m/45** (short *12.7mm akan m/45*) meaning 12.7 mm autocannon m/45.^[88] The Swedish Air Force used a different designation system which recognized the incendiary rounds as grenades, thus it was called autocannon.^[88] The AN/M2 was also produced in Finland under the export name **Colt MG 53-2**.

M296

The XM296/M296 is a further development of the AN/M2 machine gun for the OH-58 Kiowa Warrior helicopter. The M296 differs from previous remote firing variants in that it has an adjustable firing rate (500–850 rpm), while lacking a bolt latch (allowing single-shot operation).^[90] As an air-cooled gun used



.50 AN/M2 aircraft machine gun



A P-47 Thunderbolt firing its eight AN/M2 machine guns

aboard a relatively slow rotary-wing aircraft, the M296 has a burst restriction rate of 50 rounds per minute sustained fire or 150 rounds per minute maximum while conducting peacetime training requirements; the combat firing rate is unrestricted but a ten-minute cooling period after prolonged firing is mandated to avoid stoppages due to overheating.^[90]

XM213/M213, XM218, GAU-15/A, GAU-16/A, and GAU-18/A

The XM213/M213 was a modernization and adaptation of existing .50 caliber AN/M2s in inventory for use as a pintle-mounted door gun on helicopters using the M59 armament subsystem.

The GAU-15/A, formerly identified as the XM218, is a lightweight member of the M2/M3 family. The GAU-16/A was an improved GAU-15/A with modified grip and sight assemblies for similar applications. Both of these weapons were used as a part of the A/A49E-11 armament subsystem (also known as the Defensive Armament System).

The GAU-18/A is a lightweight variant of the M2/M3 and is used on the USAF's MH-53 Pave Low and HH-60 Pave Hawk helicopters. These weapons use the M2HB barrel and are typically set up as left-hand feed, right-hand charging weapons, but on the HH-60 Pave Hawks that use the EGMS (External Gun Mount System) the gun is isolated from the shooter by a recoil-absorbing cradle and all weapons are set up as right-hand charge but vary between left- and right-hand feed depending on what side of the aircraft it is on. A feed chute adapter is attached to the left- or right-hand feed pawl bracket allowing the weapon to receive ammunition through a feed chute system connected to externally mounted ammunition containers holding 600 rounds each.

AN/M3, GAU-21/A, and M3P



AN/M3 quad mount on a B-52D

During World War II, a faster-firing Browning was developed for aircraft use. The AN/M3 increased the rate of fire to around 1,200 rounds per minute while firing the same round with minimal change in weight or size.^[91] The AN/M3 was used in South Korea on the F-82 Twin Mustang (the XP-82 mounted a total of 14 AN/M3 machine guns), F-86 Sabre, F-84 Thunderjet, and F-80 Shooting Star, and in Vietnam in the XM14/SUU-12/A gun pod. Two are installed in the Embraer EMB 314 Super Tucano aircraft as a fixed wing-mounted standard weapon as designated as M3W with minor

modification with reduced weight barrel, electronic box triggered from the cockpit with 250 rounds each.

The M3 series is used by the U.S. military in two versions, the M3M and M3P. The fixed, remote-firing version, the FN *M3P*, is employed on the Avenger Air Defense System and was used on the OH-58D, augmenting the XM296 .50 cal. machine gun.^[92] The FN *M3M* flexible machine gun has been adopted by the U.S. Navy under the designation GAU-21/A for use on helicopters. The GAU-21/A is also being used by the U.S. Marine Corps to upgrade from the XM-218/GAU-16 .50 cal. machine gun for the CH-53E,^[93] on the UH-1Y Venom, on the Canadian Forces' CH-146 Griffon via the INGRESS upgrade and is to be used on the CH-53K. The Air Force is looking to replace the GAU-18 on the HH-60G Pave Hawk with the GAU-21 because of its higher 1,100 rpm rate of fire, longer 10,000-round barrel life, and lower

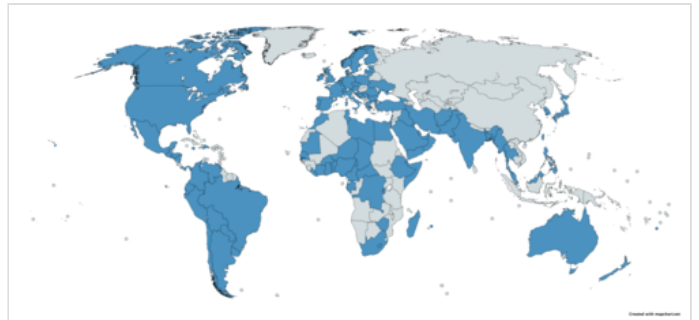
recoil through the use of a soft mount.^[94] The M3M is also the primary machine gun used by the British Royal Navy's Fleet Air Arm for helicopter armament on Wildcat and Merlin aircraft.^[95] It is also used by the German Bundeswehr on their NH-90 helicopters.^[96]

Users

The M2 family has been widely used abroad, primarily in its basic infantry configuration. A brief listing of designations for M2 family weapons follows:



A German Army ramp gunner manning
an M3M on a CH-53 Sea Stallion



Map with M2 Browning users in blue

Country	Is NATO member	Designation	Description
 Afghanistan ^[97]	No		12.7 × 99 mm Browning M2HB machine gun
 Argentina ^[98]		M2HB, AN/M3	
 Australia ^[98]	Global Partner	M2HB-QCB ^[99] (M2HB Quick Change Barrel)	
 Austria ^[98]	No	üsMG M2	
 Bahrain ^[100] : 77			
 Bangladesh ^[101]		K6 machine gun (Imported from South Korea)	
 Belgium ^[98]	Yes		
 Benin ^[98]	No		
 Bolivia ^[98]			
 Brazil ^[98]		mtr .50 M2 HB "BROWNING" and M3	
 Bulgaria ^[98]	Yes		
 Burkina Faso ^[98]	No		
 Bosnia ^[98]			
 Burundi ^[98]			
 Cameroon ^[98]			
 Canada ^[98]	Yes	FN M2HB-QCB, GAU-21	
 Central African Republic ^[102]	No		
 Chad ^[98]			
 Chile ^[98]			
 Colombia ^[98]	Global Partner		
 Cote d'Ivoire ^[98]	No		
 Croatia ^[98]	Yes		
 Democratic Republic of Congo ^[98]	No		
 Denmark ^[98]	Yes	m/50 TMG ^[103]	
		M/2001 TMG	12.7 × 99 mm FNH M2HB-QCB ^[104]
			12.7 × 99 mm FNH M3M machine gun ^[105]

 Czech Republic ^[98]			12.7 × 99 mm FNH M2HB-QCB ^[106]
 Djibouti ^[98]	No		12.7 × 99 mm Browning M2HB machine gun
 Dominican Republic ^[98]			
 Ecuador ^[98]			
 Egypt ^[98]			
 El Salvador ^[98]			
 Estonia ^[107]	Yes	<i>Browning M2</i> sometimes as <i>Raskekuulipilduja Browning M2</i>	12.7 × 99 mm Browning M2HB. Usually mounted on vehicles, such as the <u>pasi XA-180</u> and <u>XA-188</u> , but the tripod version is also in use.
 Ethiopia ^[98]	No		12.7 × 99 mm Browning M2HB machine gun
 France ^[98]	Yes	MIT 50 (Mitrailleuse cal. 50)	
 Syrian opposition ^[108]	No		
 Finland ^[109]	Yes	12,7 RSKK 2005	12.7 × 99 mm Browning M2HB machine gun in <u>protector remote weapon stations</u> in <u>Patria AMV APCs</u> . ^[110]
 Gabon ^[98]	No	12.7 × 99 mm Browning M2HB machine gun	
 Gambia ^[98]			
 Ghana ^[98]			
 Georgia ^[111]			
 Germany ^[112]	Yes	M3M, MG50	Produced by <u>FN Herstal</u> . ^[113]
 Greece ^[98]			12.7 × 99 mm Browning M2HB machine gun
 Guatemala ^[98]	No		
 Honduras ^[98]			
 India ^[98]			
 Indonesia ^[98]			
 Iran ^[98]			
 Ireland ^[98]	No	.5 Heavy Machine Gun (HMG) ^[114]	
 Israel ^[98]		0.5 כ"ק 12.7 × 99 mm M2HB-QCB, used by all ground forces (infantry, armored fighting vehicles and tanks) and naval forces	

 <u>Italy</u> ^[98]	Yes		12.7 × 99 mm Browning M2HB machine gun
 <u>Ivory Coast</u> ^[115]	No		
 <u>Jamaica</u> ^[98]			
 <u>Japan</u> ^[98]	Global Partner	12.7mm Heavy Machine Gun M2	
 <u>Jordan</u> ^[98]	No		
 <u>Katanga</u> ^[116]			
 <u>South Korea</u> ^[98]	Global Partner	K6 machine gun (standard HMG), MG50 (being phased out), M3M (used by Cheonghae Unit ^[117])	The Armed Forces received 664 M2s before the Korean War, and 4,445 were in service by the end of the war. ^[118] Later, produced locally by Yehwa Shotgun.
 <u>Kuwait</u> ^[98]	No		12.7 × 99 mm Browning M2HB machine gun
 <u>Lebanon</u> ^[98]			
 <u>Lesotho</u> ^[119]			
 <u>Liberia</u> ^[98]			
 <u>Libya</u>			12.7 × 99 mm Browning M2HB and M3 machine guns ^[120]
 <u>Latvia</u> ^[121]	Yes	M2HB-QCB	12.7 × 99 mm Browning M2HB machine gun
 <u>Lithuania</u> ^[122]			
 <u>Luxembourg</u> ^[98]		Mitrailleuse .50 M2 HB ^[123]	
 <u>Madagascar</u> ^[98]	No		
 <u>Malaysia</u> ^[98]			
 <u>Mauritania</u> ^[98]			
 <u>Mauritius</u> ^[98]			
 <u>Mexico</u> ^[98]			
 <u>Morocco</u> ^[98]			
 <u>Myanmar</u> ^[98]			
 <u>Netherlands</u> ^[98]	Yes		
 <u>New Zealand</u> ^[98]	Global Partner		
 <u>Nicaragua</u> ^[98]	No		
 <u>Niger</u> ^[98]			
 <u>Nigeria</u> ^[98]			

 Norway^[98]	Yes	12,7 mitraljøre
 Oman^[98]	No	
 Pakistan^[98]	Global Partner	
 Panama^[98]	No	
 Paraguay^[98]		
 Peru^[98]		
 Philippines^[98]		M3P ^[124]
 Poland^[125]	Yes	GAU-21
 Portugal^[98]		m/951 ^[126]
 Qatar^[98]	No	
 Romania^[98]	Yes	
 Rwanda^[98]	No	
 Saudi Arabia^[98]		
 Senegal^[98]		
 Serbia^[98]		
 Singapore^[98]		
 Slovenia^[127]	Yes	FN HERSTAL M2HB QCB Heavy Machine Gun
 Somalia^[98]	No	
 South Africa^[98]		12.7mm L4 Browning Machine Gun
 Soviet Union		M2 AA variant, <u>Lend-Lease</u> , 3100 pieces ^[128]
 Spain^{[98][129]}	Yes	
 Sweden^[130]		
 Switzerland^[98]	No	mG 64
 Sri Lanka^[98]		
 Taiwan^[98]		
 Thailand^[98]		
 Togo^[98]		
 Tonga^[98]		
 Tunisia^[98]		
 Turkey^[98]	Yes	
 Ukraine^[131]	No	

 United Arab Emirates^[98]			
 United Kingdom^[98]	Yes	L2A1	
		L6, L6A1	12.7 × 99 mm Browning M2HB machine gun; ranging gun for the L7 105 mm tank gun on the Centurion tank
		L11, L11A1	12.7 × 99 mm Browning M2HB machine gun; ranging gun
		L21A1	12.7 × 99 mm Browning M2HB machine gun; ranging gun for the 120 mm tank gun on the Chieftain tank
		L111A1 ^[132]	12.7 × 99 mm M2QCB machine gun
		M3M ^[133]	12.7 × 99 mm FN Herstal built upgraded M2 for use on Commando Helicopter Force and other units as helicopter door guns.
 United States^[98]		Browning Caliber .50 M2, M2HB, XM218/GAU-16, GAU-21	
 Uruguay^[98]	No		12.7 × 99 mm Browning M2HB machine gun
 Venezuela^[98]			
 Vietnam^[134]			
 Yemen^[98]			
 Zimbabwe^[98]			

Myths regarding M2 Browning use against personnel

It is often stated in military circles that the use of the M2 Browning against human targets is prohibited under the [Geneva Conventions](#). This is false; the Geneva Convention, contrary to popular belief, does not regulate the use of any weapon of war. It only regulates the treatment and protection of prisoners of war and non-combatants.^{[135][136]} There has been controversy surrounding the use of .50 caliber ammunition against enemy personnel, due to the explosive and incendiary [Raufoss Mk 211](#) ammunition. The [Saint Petersburg Declaration of 1868](#) states that the "military or naval" use of explosive or incendiary projectiles with a mass of under 400 g (14 oz) is forbidden by its signatory parties.^[137]

The Hague Conventions of 1899 and 1907, which are the international treaties that do regulate the use of weaponry, do not prohibit the use of .50 caliber weapons like the M2 against personnel. The US Army's own field manual concerning the employment of the M2 actually describes enemy personnel as an intended target.^[138] The origin of these myths is likely due to military commanders instructing their troops to conserve ammunition for targets that other lighter machine guns were not well suited to engage, such as aircraft or ground vehicles. One possible source of the misconception is from World War II, when American [half-track](#) units in Germany were told to stop firing their M2s at ground targets, to conserve ammunition in case of a [Luftwaffe](#) attack.^[139] Also, U.S. troops were told to use their M2s only against

enemy equipment due to shortages of ammunition during the Korean or Vietnam War.^[140] It is also possible that a restriction during the latter period limiting the use of the M40 recoilless rifle's .50-caliber spotting gun to destroy enemy equipment only, since the M40 was meant to be used against armor and firing it at personnel would give away their position before it could be used as intended, was mistakenly applied to all .50 caliber weapons.^[136]

See also

- DShK, NSV, and Kord 12.7 mm machine guns, Soviet/Russian equivalents.
- FN BRG-15 15.5 mm caliber machine gun
- HMG PK-16, 12.7×108mm Pakistani equivalent
- KPV heavy machine gun 14.5 mm caliber machine gun
- List of crew-served weapons of the U.S. Armed Forces
- List of individual weapons of the U.S. Armed Forces
- List of U.S. Army weapons by supply catalog designation
- M85 machine gun, a vehicle-borne replacement for the M2 that proved unreliable and was removed from service
- MG 131 machine gun, World War II 13 mm German aircraft-mounted gun
- MG 18 TuF, a German 13.2 mm machine gun from WWI
- Type 77/85, W85, Type 89, Type 171 12.7 mm machine guns, Chinese equivalents.

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